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In [8]: # Micro-Project 1 should use Numpy and Pandas objects to organize, clean, analyze d

import pandas as pd
import numpy as np

#has multiple sheets inside this spreadsheet... Every city has matching timestamps
dfraw = pd.read_excel(r'C:\Users\jonat\ANA_500DataMorocco.xlsx', sheet_name='Marrak

#Step 1 Query Data, see the shapes and what it Looks Like
dfraw.head(10)

# Load all sheets into a dict: {sheet_name: DataFrame}
all_sheets = pd.read_excel(r'C:\Users\jonat\ANA_500DataMorocco.xlsx', sheet_name=No

print(all_sheets.keys()) # confirm exact sheet names, e.g. dict_keys(['Table2.1',

# Set Timestamp as the index for each city's sheet, then join them side by side
Laayoune = all_sheets['Laayoune'].set_index('DateTime')
Boujdour = all_sheets['Boujdour'].set_index('DateTime')
Foum_eloued = all_sheets['Foum eloued'].set_index('DateTime')
Marrakech = all_sheets['Marrakech'].set_index('DateTime')

merged = (
    Laayoune.add_suffix('_Laayoune')
    .join(Boujdour.add_suffix('_Boujdour'), how='outer')
    .join(Foum_eloued.add_suffix('_FoumEloued'), how='outer')
    .join(Marrakech.add_suffix('_Marrakech'), how='outer')
)
merged.head(10)

dict_keys(['Laayoune', 'Boujdour', 'Foum eloued', 'Marrakech'])
```

Out[8]:

	zone1_Laayoune	zone2_Laayoune	zone3_Laayoune	zone4_Laayoune	zone5_La
DateTime					
2022-09-14 17:10:00	73.536915	120.920183	160.982021	122.236644	102
2022-09-14 17:20:00	74.403119	122.179697	159.403248	119.727278	101
2022-09-14 17:30:00	77.893555	124.701734	159.969683	118.508081	102
2022-09-14 17:40:00	78.317958	126.731194	161.969315	118.571429	102
2022-09-14 17:50:00	81.334845	125.741697	163.189366	120.688295	102
2022-09-14 18:00:00	81.885393	129.110810	166.381452	121.671637	103
2022-09-14 18:10:00	84.324060	133.028075	168.135956	121.245503	103
2022-09-14 18:20:00	88.338335	135.741555	169.904168	121.771513	105
2022-09-14 18:30:00	88.852922	134.611450	170.305196	121.639321	104
2022-09-14 18:40:00	92.165593	138.338899	172.613406	123.451568	104

In [9]: *#Step 2 Prepare: Explore and Pre-Process*

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# Checking how bad our timestamp drift is on marrakech and what our min and max is
for name, df in {'Laayoune': Laayoune, 'Boujdour': Boujdour,
                 'Foum_eloued': Foum_eloued, 'Marrakech': Marrakech}.items():
    ts = df.index
    print(name)
    print(' min:', ts.min(), ' max:', ts.max())
    print(' inferred freq:', pd.infer_freq(ts))
    print(' dtype:', ts.dtype)
    print()
```

Laayoune

min: 2022-09-14 17:10:00 max: 2024-05-24 00:00:00
inferred freq: None
dtype: datetime64[ns]

Boujdour

min: 2022-09-14 17:10:00 max: 2024-05-24 00:00:00
inferred freq: None
dtype: datetime64[ns]

Foum_eloued

min: 2022-09-14 17:10:00 max: 2024-05-24 00:00:00
inferred freq: None
dtype: datetime64[ns]

Marrakech

min: 2023-01-09 00:00:00 max: 2024-01-08 23:30:00
inferred freq: None
dtype: datetime64[ns]

```
In [13]: # checking actual typical gap between time series
for name, df in {'Laayoune': Laayoune, 'Boujdour': Boujdour,
                 'Foum_eloued': Foum_eloued, 'Marrakech': Marrakech}.items():
    diffs = df.index.to_series().diff().dropna()
    print(name)
    print(' most common interval:', diffs.mode()[0])
    print(' value counts (top 5):')
    print(diffs.value_counts().head(5))
    print(' # duplicate timestamps:', df.index.duplicated().sum())
    print()
```

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Laayoune
  most common interval: 0 days 00:10:00
  value counts (top 5):
DateTime
0 days 00:10:00      88885
-1 days +23:10:00      2
0 days 01:10:00      2
Name: count, dtype: int64
  # duplicate timestamps: 12

```

```

Boujdour
  most common interval: 0 days 00:10:00
  value counts (top 5):
DateTime
0 days 00:10:00      88885
-1 days +23:10:00      2
0 days 01:10:00      2
Name: count, dtype: int64
  # duplicate timestamps: 12

```

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Foum_eloued
  most common interval: 0 days 00:10:00
  value counts (top 5):
DateTime
0 days 00:10:00      88885
-1 days +23:10:00      2
0 days 01:10:00      2
Name: count, dtype: int64
  # duplicate timestamps: 12

```

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Marrakech
  most common interval: 0 days 00:30:00
  value counts (top 5):
DateTime
0 days 00:30:00      17488
0 days 00:00:00       3
0 days 00:16:00       2
0 days 00:14:00       2
0 days 10:00:00       1
Name: count, dtype: int64
  # duplicate timestamps: 3

```

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In [14]: # the index was not sorted from the output above, output was false, had to run sort
print(Marrakech.index.is_monotonic_increasing)

Marrakech = Marrakech.sort_index()

print(Marrakech.index.is_monotonic_increasing)

def clean_index(df):
    df = df.sort_index()
    df = df[~df.index.duplicated(keep='first')]
    return df

Laayoune = clean_index(Laayoune)

```

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Boujdour      = clean_index(Boujdour)
Foum_eloued   = clean_index(Foum_eloued)
Marrakech     = clean_index(Marrakech)

# Downsample the 10-min cities to match Marrakech's 30-min grid
Laayoune_30   = Laayoune.resample('30min').mean()
Boujdour_30   = Boujdour.resample('30min').mean()
Foum_eloued_30 = Foum_eloued.resample('30min').mean()

merged = (
    Laayoune_30.add_suffix('_Laayoune')
    .join(Boujdour_30.add_suffix('_Boujdour'), how='outer')
    .join(Foum_eloued_30.add_suffix('_FoumEloued'), how='outer')
    .join(Marrakech.add_suffix('_Marrakech'), how='outer')
)

print(merged.shape)
merged.head(10)

```

True

True

(29634, 17)

Out[14]:

	zone1_Laayoune	zone2_Laayoune	zone3_Laayoune	zone4_Laayoune	zone5_La
DateTime					
2022-09-14 17:00:00	73.970017	121.549940	160.192634	120.981961	102
2022-09-14 17:30:00	79.182119	125.724875	161.709455	119.255935	102
2022-09-14 18:00:00	84.849262	132.626813	168.140525	121.562884	104
2022-09-14 18:30:00	92.625382	139.324748	173.786221	124.403135	107
2022-09-14 19:00:00	105.361513	160.627248	200.556432	141.767547	133
2022-09-14 19:30:00	108.676651	164.539254	201.590888	145.808044	136
2022-09-14 20:00:00	109.303840	159.558735	199.641659	142.066494	136
2022-09-14 20:30:00	106.300103	158.349325	199.292568	141.001435	138
2022-09-14 21:00:00	106.177201	159.051158	197.458482	139.961136	142
2022-09-14 21:30:00	107.348845	157.539046	195.285820	139.797847	143

In [15]: *# export prepared data to a csv before step 3 analyze so we are ready to go on that*

```
merged.to_csv(r'C:\Users\jonat\OneDrive\Desktop\NU Homework\ANA500\ANA500Project\AN
```

In []: